

Datatypes

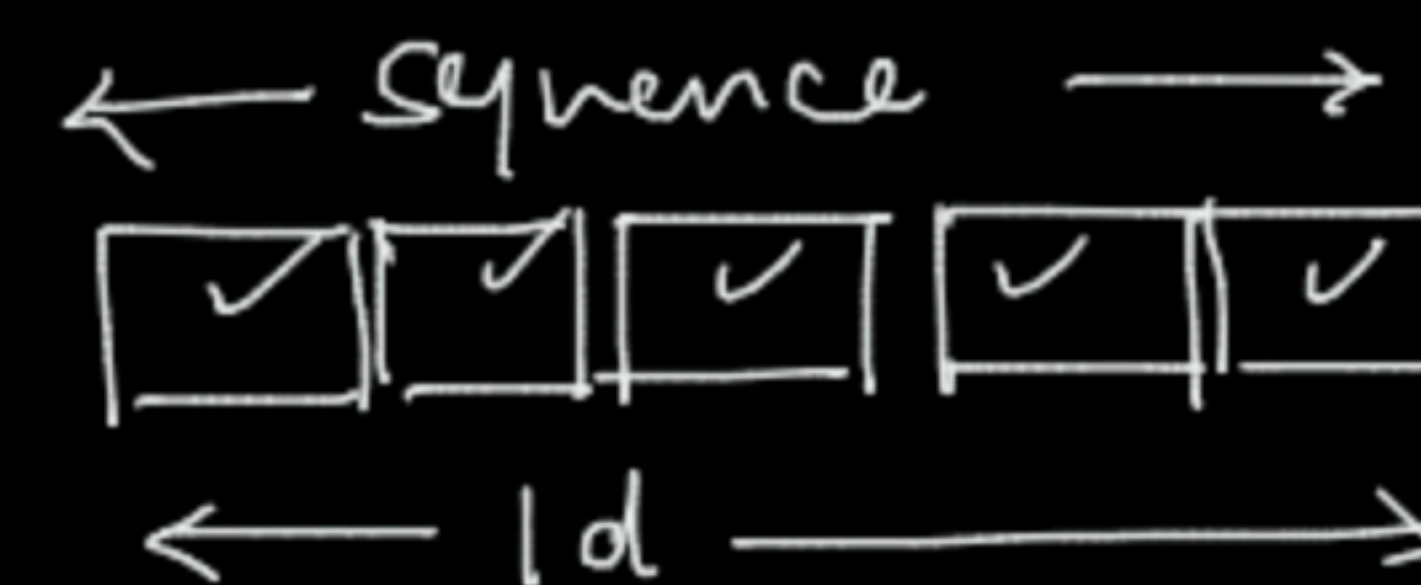
Basic Types

Numeric — { integral — int, non integral — float }

Boolean → bool

5.2 → float
x → float

Collections — { Sequence — { Mutable — list, Immutable — tuple, str, dict, set, frozenset }, Mapping — dict { k1: v1, k2: v2 }, Set — { e1, e2, e3, ... } }



Pandas — { Series ⇒ Sequence (1d), DataFrame ⇒ Multiple Series (2d) }

eid	name			→ attribute
100	Tom	—	—	} actual
200	Jim	—	—	
300	Tom	—	—	
400	Jim	—	—	

Modules, packages, libraries

Module: Collection of Variables, Functions, Classes.

Template

list → class

↳ l1 ✓
 ↳ l1 ✓
 ↳ shop_l1 ✓
 } object of type list
 Methods

l1.append()

tuple → class

↳ t1 ✓
 ↳ my_tup ✓
 ↳ shop_tup ✓
 } objects of type tuple

index() ✓
count() ✓

t1.append() x

Core Python

type(l1) → list

l1 = [1, 2, 3]

l1.append()

l1 = List(-)
= [, ,]

→ Pandas

→ Series

s1 = Series(-)
↳ info()
↳ head()

s1.head()

d1 = DataFrame(- - -)

DataFrame ← d1.join() - [join()
sort()
d1.sort()

1. Create an object of type list/tuple/str
2. Call any method on the object

module math.py

$\pi = 3.142$

def log(x): ✓

==

return

def exp(x):

==

return

def cos(x):

==

return

def sin(x):

==

Write nice

P1.py →

import math

$y = \text{math.log}(x) +$
 $2 \text{math.sin}(x)$

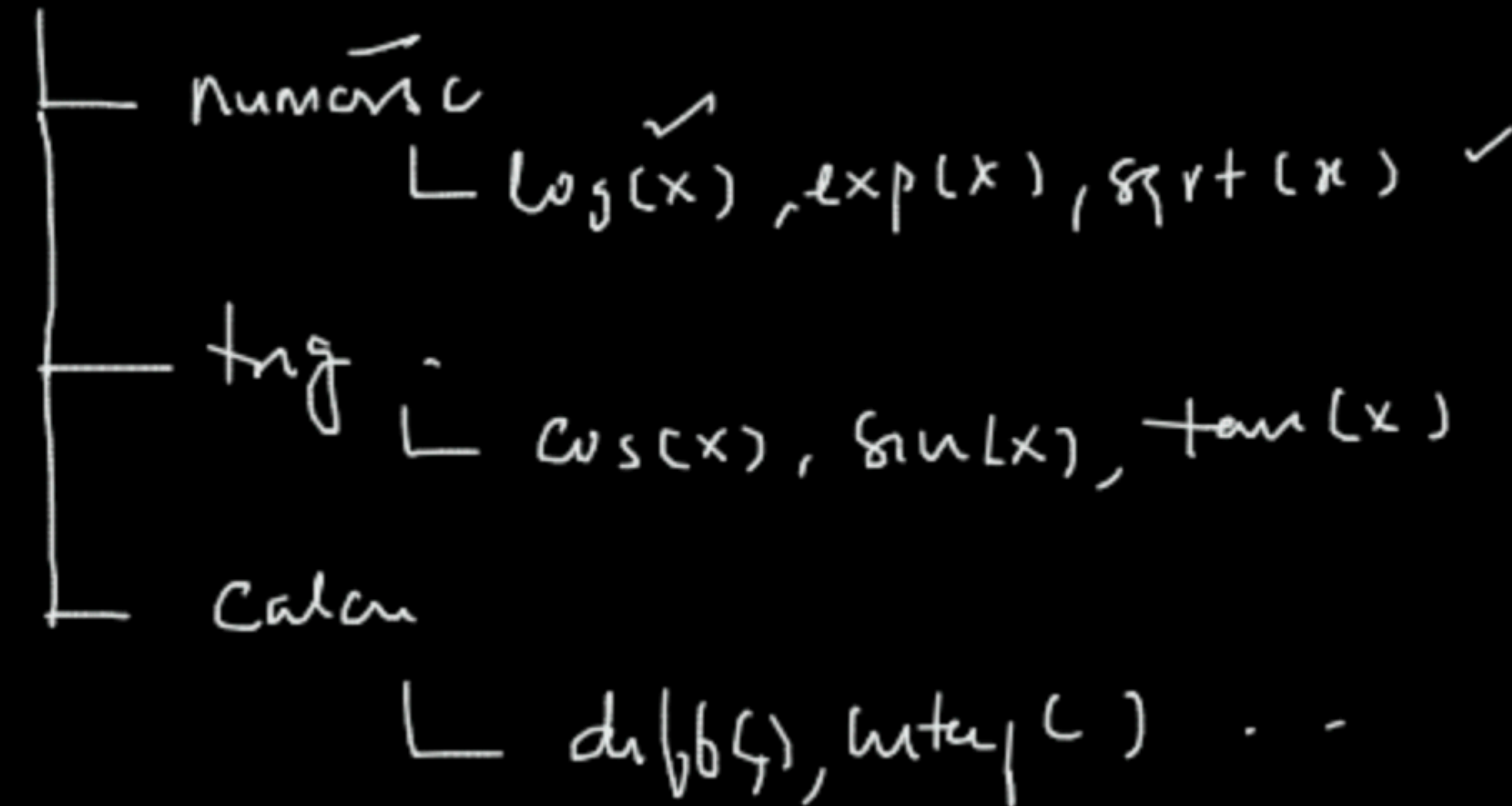
P2.py

import math

$y = \text{math.log}(x) +$
 $\text{math.sin}(x)$

Naming Conventions

math



from math import numeric

$y = \text{numeric.log}(x)$

Classes → MyClass (title case)
→ Evaluation

Variables, fns, modules
→ snake_case

from math.numeric import sqrt
↓
fn

import pandas, numpy
Numpy x

Pandas ✓

↳ Fns → read_csv() ✓, merge() ✓ Fns

Classes → Series — methods() ✓

DataFrame → methods()

sl = Series()

sl.info()

Pandas.read_csv()

↳ DataFrame
e df →

id	ename	job	salry
—	—	—	—
—	—	—	—
—	—	—	—

↓
Series

↓
Series

↓
Series

↓
Series

→ DataFrame

edf =>

		0	1	2	3	Columns
		Id	ENAME	Job	Salary	
0	E1	100	Thomas	Mgr	2500	
1	E2	200	Mary	pgmr	3400	
2	E3	300	Kate	HR	4700	
3	E4	400	Tim	Mgr	3200	
4	E5	500	Jim	pgmr	6400	
5	E6	600	William	HR	2000	

index

[

[100, Thm, Mr, 2500],

[200, m. . .] ,

[.]

]